Cheatsheet AZ-104 — 16 · Backup and recovery: vaults, policies, restore and Site Recovery

1. The two vaults

	Recovery Services vault	Backup vault
What it protects	**Azure VMs**, **Azure Files** (SMB), **SQL Server and SAP HANA in VMs**, **MARS** (Windows files, folders and system state), **MABS / DPM** (on-prem), and **Azure Site Recovery**	**Managed disks**, **Blobs** (operational and vaulted), **Azure Database for PostgreSQL / MySQL**, **AKS**
Region	For **backup**, the vault must be in the **same region** as what it protects; for **Site Recovery**, in **any region except the source one**	Same region as the resource
Redundancy	**LRS / ZRS / GRS** (with optional **Cross Region Restore**); set **before the first backup**	LRS / ZRS / GRS
Security	**Soft delete** (14 days by default, free, **always on**, extendable to 180), **MUA** (multi-user authorization with **Resource Guard** for critical operations), vault **immutability**, CMK, private endpoints	Same
Management	**Business Continuity Center** (formerly Backup center): single view of vaults, jobs, policies, alerts and reports across all subscriptions	Same

- "Vault in another region to protect against regional disaster" → that is **GRS + Cross Region Restore** (backup) or **Site Recovery** (DR); VM backup requires the vault in its region.

2. Azure VM backup

Aspect	Detail
How it works	The **VMSnapshot** extension (Windows: **VSS**, app-consistent; Linux: file-consistent, app-consistent with **pre/post scripts**) takes a **snapshot** (instant tier, in the RG `AzureBackupRG_<region>_1`) and then transfers it **incrementally** to the vault (can take up to 8 h); there is also the **agentless crash-consistent** mode
Standard policy	Once a **day or a week**; snapshot retained **1-5 days** (instant restore); daily / weekly / monthly / yearly retention (up to 9,999 days)
Enhanced policy	**Several per day** (every 4, 6, 8 or 12 h), snapshot up to **30 days**, **zone-redundant** snapshots; **mandatory** for **Premium SSD v2** / **Ultra** disks and confidential VMs; for **Trusted Launch** it is the portal's option (Standard only via recent CLI / PowerShell / REST)
Restore	**Create new VM** (new, same or different RG / VNet) · **Restore disks** (to a storage account / RG; then the VM is created manually or with a template; to change settings) · **Replace existing** (replaces the disks of the original VM; keeps NIC and IP) · **Cross Region Restore** (in the paired region; requires **GRS + CRR** enabled) · **Cross subscription restore** · **File recovery** (script that mounts the recovery point as an iSCSI disk for 12 h to copy individual files) · restore to a specific **zone**; **selective disk** backup / restore
Stop	**Stop backup**: *retain backup data* (no more copies are taken, restores still possible) or *delete backup data* (goes into **soft delete** for 14 days)
Requirements	VM and vault in the **same region**; **VM agent** installed; disks up to **64 TB** (32 disks, 512 TB in total); VMs with **ADE** need Backup to read the keys from the Key Vault (BEK / KEK copy); no **CanNotDelete locks** on the snapshot RG; port 443 to the Storage / Backup service tags
Cost	Per **protected instance** (actual data size, excluding the temporary disk) + vault storage; **vault-archive tier** for long retention

- "Backup every 4 hours" → Enhanced. "Restore to a different VNet / change size" → Restore disks + create VM. "Restore a single file" → File recovery. "Backup fails after enabling encryption" → Backup permissions on Key Vault. "Backup fails: lock" → remove the lock from `AzureBackupRG`.

3. Other workloads

Workload	How	Exam details
Azure Files	Share **snapshots** orchestrated by Backup, stored in the **same storage account**; daily or **hourly** policy (several per day); GFS retention	Restore of the **whole share** or **item-level** (files / folders) in place or to another location; Backup enables soft delete and places a **lock** on the account; needs the account and the vault in the same region
SQL Server / SAP HANA in VM	Workload extension in the VM; full, differential and **log every 15 min**	Restores **individual databases** to a point in time
MARS agent (Microsoft Azure Recovery Services)	Agent on **Windows** (on-prem or VM) for **files, folders and system state**; registered with the **vault credentials** and a **passphrase** (if lost, no restore)	Maximum **3 backups per day**; **no Linux**, no full VM, not app-aware; bandwidth throttling; restores to the same machine or another one
MABS / DPM	On-prem backup server (Hyper-V, VMware, SQL, Exchange, SharePoint) with disk → vault copy	"back up on-premises VMware VMs to Azure" → MABS
Disks (Backup vault)	Operational incremental snapshots, agentless	For disks without a protected VM
Blobs (Backup vault)	**Operational**: continuous, uses versioning + change feed + soft delete + **point-in-time restore** in the account itself · **Vaulted**: copy in the vault	"restore the container to yesterday" → operational

4. Monitoring backup

Feature	Detail
Jobs	Status of backups and restores per vault (and centralized in Business Continuity Center)
Azure Monitor alerts for backup	**Built-in** alerts (backup / restore failure, security events such as deletions) already active in the vault; routed with **action groups** (email, SMS, runbook); shown in Monitor → Alerts
Backup reports	Vault diagnostic setting → **Log Analytics** (`AddonAzureBackup*` categories); workbooks with compliance, consumption, jobs; email reports

Feature	Detail
Metrics	Backup Health Events, Restore Health Events (metric alerts)

5. Azure Site Recovery (disaster recovery)

Piece	Detail
Scenarios	**Azure** → **Azure** (another region), **zone** → **zone**, **VMware / physical** → **Azure** (replication appliance), **Hyper-V** → **Azure** (provider + agent, with or without VMM)
A2A components	Vault in a **different region** from the VMs; **cache storage account** in the **source** region; **Mobility service** extension (installs itself); target resources created with the **asr** suffix (RG, VNet, availability set); `-ASRReplica` replica disks
Replication	**Continuous**: every write goes to the cache and from there to the target; **crash-consistent points every 5 min** (fixed); **app-consistent** per the policy (default **disabled**, 0 h); **default recovery point retention 1 day** (extendable, e.g. up to 15); **multi-VM consistency** (replication group) for shared points
Network mapping	Which target VNet / subnet corresponds to each source VNet; target IPs and NSG configurable per VM
Test failover	Brings up the VMs in an **isolated** test network **without touching** production or the replication; for drills ("perform a DR drill without impact")
Failover	Choose the point: *latest* (processes pending data), *latest processed* (minimum RTO), *latest app-consistent*, or custom; creates the VMs at the target; then **Commit** (deletes the other points)
Re-protect	Reverses the replication (target → source) to be able to go back
Failback	Failover back to the source once re-protected
Recovery plan	Groups VMs, defines **boot order** by groups, **scripts** (Automation runbooks) and **manual actions**; one click for the whole tier
Requirements	Managed disks; outbound 443 to the Storage, AzureSiteRecovery, EventHub, AzureActiveDirectory service tags; capacity at the target; target VM size configurable
Cost	Per protected instance (first 31 days free) + cache and replica storage

- **Backup vs Site Recovery:** backup = point-in-time copies to recover data or lost VMs (RPO of hours, high RTO); ASR = continuous replica to keep operating in another region (RPO of minutes, RTO of minutes). "RPO 1 hour, RTO 2 hours" → ASR. "Keep 7 years of recovery points" → backup. Both can coexist, but when **moving** a VM both must be disabled.

6. Commands that show up

Task	PowerShell (Az.RecoveryServices)	Azure CLI
Vault	`New-AzRecoveryServicesVault -Name rsv1 -ResourceGroupName RG1 -Location westeurope` · `Set-AzRecoveryServicesVaultContext -Vault $v` (or `-VaultId` on each cmdlet) · `Set-AzRecoveryServicesBackupProperty -Vault $v -BackupStorageRedundancy GeoRedundant` · CRR: `Set-AzRecoveryServicesBackupProperty -EnableCrossRegionRestore`	`az backup vault create -n rsv1 -g RG1 -l westeurope` · `az backup vault backup-properties set -n rsv1 -g RG1 --backup-storage-redundancy GeoRedundant --cross-region-restore-flag true` · `az backup vault backup-properties set --soft-delete-feature-state Enable`
Policy	`Get-AzRecoveryServicesBackupProtectionPolicy -Name DefaultPolicy` · `$sched = Get-AzRecoveryServicesBackupSchedulePolicyObject -WorkloadType AzureVM -PolicySubType Enhanced; $ret = Get-AzRecoveryServicesBackupRetentionPolicyObject -WorkloadType AzureVM; New-AzRecoveryServicesBackupProtectionPolicy -Name p1 -WorkloadType AzureVM -SchedulePolicy $sched -RetentionPolicy $ret`	`az backup policy list -v rsv1 -g RG1` · `az backup policy create -n p1 -v rsv1 -g RG1 --backup-management-type AzureIaasVM --policy @policy.json` · `az backup policy set`
Protect VM	`Enable-AzRecoveryServicesBackupProtection -Policy $pol -Name vm1 -ResourceGroupName RG1` · `Backup-AzRecoveryServicesBackupItem -Item $item` (now)	`az backup protection enable-for-vm -v rsv1 -g RG1 --vm vm1 --policy-name DefaultPolicy` · `az backup protection backup-now -v rsv1 -g RG1 -c vm1 -i vm1 --backup-management-type AzureIaasVM --retain-until 31-12-2026`
Restore	`$rp = Get-AzRecoveryServicesBackupRecoveryPoint -Item $item; Restore-AzRecoveryServicesBackupItem -RecoveryPoint $rp[0] -StorageAccountName st1 -StorageAccountResourceGroupName RG1 -TargetResourceGroupName RG-restore` (restore disks) · `-RestoreAsUnmanagedDisks` · cross region: `-RestoreToSecondaryRegion`	`az backup recoverypoint list -v rsv1 -g RG1 -c vm1 -i vm1 --backup-management-type AzureIaasVM` · `az backup restore restore-disks -v rsv1 -g RG1 -c vm1 -i vm1 --rp-name <rp> --storage-account st1 --target-resource-group RG-restore` · `--restore-mode OriginalLocation` (replace existing) · `az backup restore files mount-rp` (file recovery)
Stop / jobs	`Disable-AzRecoveryServicesBackupProtection -Item $item [-RemoveRecoveryPoints -Force]` · `Get-AzRecoveryServicesBackupJob -Status InProgress` · `Undo-AzRecoveryServicesBackupItemDeletion` (soft delete)	`az backup protection disable -v rsv1 -g RG1 -c vm1 -i vm1 [--delete-backup-data true]` · `az backup job list -v rsv1 -g RG1 -o table` · `az backup protection undelete`
Files and MARS	`Enable-AzRecoveryServicesBackupProtection -WorkloadType AzureFiles -Name share1 -StorageAccountName st1 -Policy $pol` · MARS: download the agent + `Get-AzRecoveryServicesVaultSettingsFile -Backup` (vault credentials)	`az backup protection enable-for-azurefileshare -v rsv1 -g RG1 --storage-account st1 --azure-file-share share1 --policy-name p1`
Site Recovery	`New-AzRecoveryServicesAsrFabric`, `New-AzRecoveryServicesAsrProtectionContainerMapping`, `New-AzRecoveryServicesAsrPolicy -AzureToAzure -RecoveryPointRetentionInHours 24 -ApplicationConsistentSnapshotFrequencyInHours 4`, `New-AzRecoveryServicesAsrReplicationProtectedItem`, `Start-AzRecoveryServicesAsrTestFailoverJob`, `Start-AzRecoveryServicesAsrUnplannedFailoverJob -Direction PrimaryToRecovery`, `Start-AzRecoveryServicesAsrCommitFailoverJob`, `Update-AzRecoveryServicesAsrProtectionDirection` (re-protect)	No native ASR commands in Azure CLI (portal / PowerShell / REST)

7. Decision tree

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What does the wording ask for?
├─ Protect Azure VMs against deletion / corruption ..... Recovery Services vault + policy (Standard / Enhanced)
├─ Several backups a day, Premium SSD v2 or Trusted Launch ..... Enhanced policy
├─ Restore a file from a VM ..... File recovery (iSCSI mount)
├─ Restore the VM with another size / network ..... Restore disks + create VM
├─ Restore in the paired region ..... GRS + Cross Region Restore
├─ Protect against backup deletion / ransomware ..... Soft delete (14–180 days) + MUA + immutability
├─ Files and folders of an on-prem Windows server ..... MARS agent (3 per day)
├─ On-prem Hyper-V / VMware VMs to Azure (backup) ..... MABS / DPM
├─ Azure Files share with per-file restore ..... Azure Files backup (snapshots)
├─ SQL databases in a VM with 15 min RPO ..... SQL in VM backup (workload)
├─ Disks / blobs / PostgreSQL ..... Backup vault
├─ Keep operating in another region if the primary fails (RTO minutes) ... Site Recovery A2A + recovery plan
├─ Test DR without affecting production ..... Test failover
├─ Return to the original region after DR ..... Re-protect + failover (failback)
├─ Notify when a backup fails ..... Built-in Backup alerts + action group
└─ Backup compliance report ..... Backup reports (Log Analytics)
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8. Signals → answer

Wording	Answer
"vault must be in a different region than the VMs"	For **backup** it is impossible (same region); for **ASR** yes
"change vault redundancy to GRS" fails	Items are already protected: it is set before the first backup
"restore the VM in the paired region"	Cross Region Restore (GRS + CRR enabled beforehand)
"backup twice a day"	Enhanced policy
"Trusted Launch VM, backup from the portal"	Enhanced policy (Standard only via recent CLI / PowerShell / REST)
"recover a deleted backup"	Soft delete: Undelete within 14 days (up to 180 if extended)
"prevent admins from deleting backups"	MUA with Resource Guard + immutability
"backup fails: CanNotDelete lock"	Remove the lock from the `AzureBackupRG_*` RG
"VM with Azure Disk Encryption backup fails"	Grant Backup permissions on Key Vault (Backup Management Service)
"restore only one disk"	Selective disk restore
"restore files without restoring the VM"	File recovery script (iSCSI, 12 h)
"MARS on Linux"	Does not exist: MARS is Windows only (use VM backup or MABS)
"MARS backs up 5 times a day"	Maximum 3
"restore a single file from the Azure file share"	Item-level restore from the snapshot
"on-premises VMware VMs replicated to Azure"	Site Recovery (appliance)
"RPO of minutes, RTO of minutes"	Site Recovery
"test the DR plan without downtime"	Test failover (isolated network)
"VMs must start in order (DB before web)"	Recovery plan with groups
"after failover, replicate back"	Re-protect
"Mobility service not installed"	Installs itself when A2A replication is enabled (check outbound 443)
"app-consistent recovery points every 4 hours"	Replication policy (app-consistent frequency; disabled by default)

9. Mistakes to avoid

1. Creating the backup vault in a different region from the VMs (the opposite of Site Recovery).
2. Changing the vault redundancy after the first backup: it cannot be done.
3. Using a Standard policy with Premium SSD v2 / Ultra (or with Trusted Launch from the portal).
4. Confusing backup (points in time, retention) with Site Recovery (continuous replica, failover).
5. Performing a real failover to "test" DR: that is Test failover.
6. Expecting MARS on Linux or more than 3 daily copies; or using MARS for a full VM.
7. Forgetting soft delete when working out when the data disappears (14 more days).
8. Placing locks on the Backup snapshot resource group.
9. Moving a VM to another subscription / RG without stopping backup and replication first.